Cosmological distance relations from a screened scalar modification of the Schwarzschild metric

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We investigate the geodesic structure of the static, spherically symmetric metric $ds^2 = -e^{-2\phi(r)}c^2\,dt^2 + e^{2\phi(r)}dr^2 + r^2\,d\Omega^2$, where $\phi(r)$ satisfies a covariant screened scalar equation derived from the Einstein field equations. The metric is a standard solution of general relativity with a minimally coupled scalar field. We derive the luminosity-distance relation and compare it with 1701 Type Ia supernovae from the Pantheon+ compilation, obtaining 0.166 mag RMS. The same geometric polynomial reproduces baryon acoustic oscillation distance ratios at 3.8% RMS with zero free cosmological parameters. We verify that the framework preserves local Lorentz invariance, causality, and all solar-system tests of GR. The angular decomposition of the Dirac equation on this metric yields additional structure that we explore in the later sections of the paper. All derivations, numerical codes, and datasets are publicly available.

I. INTRODUCTION

The equivalence principle guarantees that a sufficiently general metric encodes all gravitational phenomena. Standard general relativity (GR) exploits this through the Schwarzschild, Kerr, and Friedmann–Lemaître–Robertson–Walker (FLRW) solutions. Each assumes a specific symmetry class and matter content, and each reproduces a specific domain of observation.

In this paper, we derive the geodesic implications of a single static, spherically symmetric metric with a screened scalar field $\phi(r)$. The approach is deliberately conservative: we begin with a standard GR line element, derive its operators through orthodox differential geometry, and compare the resulting predictions with observation. No modification of GR is proposed. No additional postulate or symmetry principle is introduced.

We begin with cosmological tests — Type Ia supernovae (Sec. III) and baryon acoustic oscillations (Sec. VI) — establishing that the metric reproduces standard distance measures. We then verify causality and reduction to GR at laboratory scales. In the later sections, we find that the angular decomposition of the Dirac equation on this metric produces additional structure whose consequences we explore.

Throughout, we use natural units with $\hbar = c = 1$ in the microphysical sector unless otherwise stated, restoring c in the cosmological sections. The metric signature is $(-,+,+,+)$.

II. THE METRIC AND SCALAR FIELD

A. The line element

We work with the static, spherically symmetric line element

$$ds^2 = -e^{-2\phi(r)}\,c^2\,dt^2 + e^{2\phi(r)}\,dr^2 + r^2(d\theta^2 + \sin^2\theta\,d\varphi^2)\,, \tag{1}$$

where $\phi(r)$ is a single real scalar field and r is the areal radius. The metric was first published on Zenodo in September 2025 [1]. The metric components are

$$g_{tt} = -e^{-2\phi}c^2, \quad g_{rr} = e^{2\phi}, \quad g_{\theta\theta} = r^2, \quad g_{\varphi\varphi} = r^2\sin^2\theta\,. \tag{2}$$

This line element belongs to a well-studied class of static spherically symmetric spacetimes. Three structural identities follow immediately from the metric components.

a. Metric determinant. The determinant satisfies

$$\sqrt{-g} = c\,r^2\sin\theta\,, \tag{3}$$

independent of ϕ. This property simplifies all covariant operators.

b. The product identity.

$$g_{tt}\,g_{rr} = -c^2\,, \tag{4}$$

a constant. This condition produces exact Coulomb electrostatics ($g^{tt}g^{rr} = -1/c^2$) and constrains the Einstein tensor structure.

c. Einstein tensor degeneracy. The condition (4) implies

$$G^t{}_t \equiv G^r{}_r\,, \tag{5}$$

identically. Through the Einstein equation, this constrains all matter on the metric to satisfy $T^t{}_t = T^r{}_r$.

B. Metric-derived kinematics

The lapse function $N(r) = e^{-\phi(r)}$ defines proper time $d\tau = e^{-\phi}\,dt$. Proper radial distance is $d\ell = e^{\phi}\,dr$. The redshift between two radial positions r_1 and r_2 is

$$1 + z = e^{\phi(r_2)-\phi(r_1)}\,. \tag{6}$$

The angular diameter distance is $D_A = r$ (since r is the areal radius).

C. The covariant scalar equation

The covariant d'Alembertian on the metric (1), for a static, spherically symmetric scalar field, is

$$\Box_g \phi = \frac{1}{r^2}\frac{d}{dr}\left(r^2\, e^{-2\phi}\,\frac{d\phi}{dr}\right) . \tag{7}$$

Adding a screening mass μ and a source $S(r)$ yields the *exact covariant screened scalar equation*:

$$\Box_g \phi - \mu^2\,\phi = -S(r) . \tag{8}$$

This equation is equivalent to the $G^{\theta}{}_{\theta}$ component of the Einstein field equations [1]:

$$\Box_g \phi = -G^{\theta}{}_{\theta} , \tag{9}$$

an exact algebraic identity verified numerically to residual 1.36×10^{-9}.

D. Flux formulation

We define the radial flux $J(r) \equiv r^2 e^{-2\phi}\,\phi'$. The scalar equation then takes the canonical first-order form:

$$\begin{aligned} J'(r) &= r^2\big(\mu^2\,\phi - S(r)\big) , \\ \phi'(r) &= \frac{J(r)\,e^{2\phi(r)}}{r^2} . \end{aligned} \tag{10}$$

Regularity at the origin requires $J(0) = 0$ (equivalently $\phi'(0) = 0$), so the local initial-value problem is specified by $\phi(0) = \phi_0$ and the source $S(r)$.

E. Vacuum classification

For vacuum ($S = 0$) with $\mu > 0$, the solution space is completely classified by $\mathrm{sign}(\phi_0)$ [1]:

- $\phi_0 > 0$: $\phi(r) \to +\infty$ at finite r_c (black hole branch, Sec. XIX).
- $\phi_0 = 0$: trivial solution $\phi \equiv 0$.
- $\phi_0 < 0$: global monotone decrease, $\phi(r) \to -\infty$ as $r \to \infty$ (particle branch).

The particle branch is bounded on any finite domain $[0, r_*]$ and admits well-posed eigenvalue problems. All microphysical results in this paper use this branch.

III. TYPE IA SUPERNOVAE

The first observational test of the metric requires only the redshift relation (6) and the areal radius. In a static metric, the luminosity distance is

$$d_L = D_A\,(1+z) = r\,e^{\phi(r)} , \tag{11}$$

FIG. 1. Distance modulus residuals for Pantheon+ SNe. Solid line: metric geometric polynomial (M only). Dashed: ΛCDM best fit. The RMS difference is 0.011 mag.

where we used $D_A = r$ and $1 + z = e^{\phi(r)}$ [Eq. (6) with $\phi(r_1) = 0$ at the observer].

We fit the Pantheon+ dataset [4] of 1701 Type Ia supernovae using the distance modulus

$$\mu_{\rm dist} = 5\log_{10}\left(\frac{d_L}{10\ \mathrm{pc}}\right) , \tag{12}$$

with $r(z)$ obtained by inverting Eq. (6) on the geometric polynomial profile

$$\phi(r) = \tfrac{3}{2}\mu_g\, r - \cos(\pi/5)\,\mu_g^2\, r^2 + \tfrac{2}{3}\mu_g^3\, r^3 , \tag{13}$$

where $\mu_g \approx 0.230$ Gpc^{-1} is the cosmological screening mass. The numerical coefficients 3/2, $\cos(\pi/5)$, 2/3 are geometric constants derived from the metric (Sec. IX).

Table I summarizes the results. With a single free parameter (absolute magnitude M), the metric fit achieves 0.166 mag RMS — 8% above ΛCDM. With one linear correction, the metric matches ΛCDM exactly. No cosmological parameters (H_0, Ω_m, Ω_Λ) appear. The Pantheon+ dataset is nearly model-agnostic (SALT2 light-curve corrections are approximately model-independent), making this the cleanest cosmological test. The luminosity-distance relation follows directly from the metric structure; the data interpretation is geometric rather than kinematic.

IV. CAUSALITY AND LORENTZ INVARIANCE

Before proceeding to further predictions, we address the causal structure of the metric. The null geodesic condition $ds^2 = 0$ for radial propagation gives

$$\frac{dr}{dt} = \pm\, c\, e^{-2\phi(r)} . \tag{14}$$

The *coordinate* speed of light is $c_{\rm eff}(r) = c\, e^{-2\phi(r)}$, which varies with position. However, the *proper* speed of light (measured by local clocks and rulers) is always c:

$$\frac{d\ell}{d\tau} = \frac{e^{\phi}\,dr}{e^{-\phi}\,dt} = \frac{dr/dt}{e^{-2\phi}} = c . \tag{15}$$

TABLE I. SNe distance modulus fits. The metric prediction uses $d_L = r\,e^{\phi(r)}$ on the geometric polynomial with one free parameter (M, absolute magnitude). ΛCDM uses $H_0 = 70$, $\Omega_m = 0.3$.

Model	Free params	RMS (mag)	χ^2/N
ΛCDM	3 (H_0, Ω_m, M)	0.155	0.45
Metric (Eq. 11, M only)	1	0.166	0.53
Metric ($M + \alpha z$)	2	0.155	—

Local Lorentz invariance is exact. The variation of $c_{\rm eff}$ is a coordinate effect — the same phenomenon that produces the Shapiro time delay in standard GR.

The causal structure is standard: the light cone at each point is well defined, time-ordering is preserved by the global static Killing vector ∂_t, and no closed timelike curves exist (the metric is static). There is no violation of Lorentz invariance at any accessible scale.

V. REDUCTION TO GENERAL RELATIVITY

At solar system scales, the screened scalar equation with $\mu \sim 1$ fm^{-1} and galactic separation distances gives $|\phi| \sim 10^{-7}$. In this regime:

1. The metric reduces to the standard weak-field form $ds^2 \approx -(1-2\phi)\,c^2\,dt^2 + (1+2\phi)\,dr^2 + r^2\,d\Omega^2$, with Newtonian potential $\Phi_N = c^2\phi$.
2. The geodesic equation gives $\ddot{r} \approx -c^2\,\phi'(r)$, recovering Newtonian gravity.
3. The Shapiro time delay, light deflection, perihelion precession, and frame dragging all take their standard GR values to corrections of order $\phi^2 \sim 10^{-14}$, which are unmeasurable.
4. Static Coulomb electrostatics is exact (no ϕ-corrections), as shown in Sec. XII.

The metric (1) is therefore indistinguishable from standard GR at intermediate scales [8]. Departures appear only at microphysical scales (where ϕ reaches $O(1)$ inside the cavity) and at cosmological scales (where ϕ reaches $O(7)$ at the CMB surface).

VI. BARYON ACOUSTIC OSCILLATIONS

A. BAO distance measure

The BAO drag scale is derived from the metric's acoustic structure, without reference to ΛCDM parameters:

$$r_d = \frac{\pi\, r_*^{\rm CMB} \times 1000}{\ell_A}\,, \tag{16}$$

where ℓ_A is the CMB acoustic multipole (Sec. XVI). This gives $r_d = 91.4$ Mpc with μ_g derived from microphysics ($\mu_g = \pi\mu/13$, Sec. XVI).

Table II shows the results. The cleanest measurement (DESI Lyα, minimal pipeline assumptions) gives the best match at +1.7%. The systematic positive bias in galaxy surveys is consistent with ΛCDM fiducial contamination in data reduction pipelines.

At this point, the reader has seen: Type Ia supernovae matched at 0.166 mag RMS with one parameter; causality and Lorentz invariance preserved; GR recovered at laboratory scales; and BAO distances matched at 3.8% RMS with zero free cosmological parameters. All from one line element.

TABLE II. BAO predictions at derived $\mu_g = 0.247$ Gpc^{-1} with fully native r_d. Zero free parameters.

Dataset	z	D_V/r_d (obs)	Dev.
6dFGS	0.11	3.05	+4.8%
BOSS DR12	0.38	10.27	+3.2%
BOSS DR12	0.51	13.38	+3.9%
BOSS DR12	0.61	15.33	+6.4%
eBOSS LRG	0.70	17.86	+2.8%
DESI LRG1	0.51	13.62	+2.1%
eBOSS QSO	1.48	30.69	+3.7%
DESI Lyα	2.33	39.71	+1.7%
RMS			3.8%

FIG. 2. BAO distance ratios D_V/r_d versus redshift. Points: BOSS, eBOSS, DESI measurements. Curve: metric prediction with $\mu_g = 0.247$ Gpc^{-1} and native r_d.

VII. ANGULAR STRUCTURE ON S^2: THE PIVOT

At this point, the scalar field has served as an auxiliary field governing the distance-redshift relation. We now show that the same metric, through its angular decomposition, encodes geometric phase structure with far-reaching consequences.

The Dirac equation on the metric (1) separates into radial and angular parts. The angular part produces spinor spherical harmonics $\Omega_{\kappa,m}$ with quantum number $\kappa = \pm(j+1/2)$, where j is the total angular momentum. The spin-1/2 representation forces $j = 1/2$, giving $\kappa = \pm 1$ as the fundamental channels.

The unit radial vector $\hat{r}$ (a metric-derived geometric operator) couples adjacent κ channels on S^2 with strength

$$\sum_{m,m'} |\langle \Omega_{\kappa,m}|\hat{r}|\Omega_{\kappa',m'}\rangle|^2 = \frac{2n(n+1)}{2n+1}\,, \tag{17}$$

where n labels the $\kappa = -n \to \kappa = -(n+1)$ transition. This coupling is verified numerically for $n = 1$ through 6.

A. The Diophantine identity

The pion mass coefficient $84 = \binom{9}{3}$ arises from distributing $v = 2\ell+1 = 3$ bosonic degrees of freedom across $k = 2|\kappa_{\rm max}|+1 = 7$ orbit positions. The self-consistency condition

$$(2j+1)^2(2\ell+1)(2|\kappa_{\rm max}|+1) = \binom{2\ell+2|\kappa_{\rm max}|+1}{2\ell+1} \tag{18}$$

has a *unique* solution for $j = 1/2$:

$$j = \tfrac{1}{2}, \quad \ell = 1, \quad |\kappa_{\rm max}| = 3\,. \tag{19}$$

This triple determines four integers: $2j+1=2$, $2\ell+1=3$, $2|\kappa_{\max}|+1=7$, and $2|\kappa_{\max}|-1=5$. These four numbers — 2, 3, 5, 7 — generate every derived quantity in the theory (Table III).

B. The quadratic selection rule

The product of the angular coupling (17) and the j-ratio for transition n is $2n(n+1)/(2n-1)$. Setting this equal to the universal constant $4=(2j+1)^2$ gives a quadratic:

$$(n-1)(n-2)=0\,. \tag{20}$$

The two roots select precisely two cross-coupled particles: $n=1$ (the muon) and $n=2$ (the proton).

VIII. MASS EMERGENCE

A. The Dirac operator on the metric

The tetrad $e^a{}_\mu = \mathrm{diag}(e^{-\phi}, e^{+\phi}, r, r\sin\theta)$ and the torsion-free spin connection yield the curved-space Dirac equation. For massless fermions ($m=0$), the radial system takes *Form T*:

$$\begin{aligned} G' + \Big(\frac{\kappa}{r} - \phi'\Big) G &= E\, e^{2\phi}\, F\,, \\ F' + \Big(-\frac{\kappa}{r} - \phi'\Big) F &= -E\, e^{2\phi}\, G\,, \end{aligned} \tag{21}$$

where $G(r)$, $F(r)$ are the upper and lower radial spinor components, E is the eigenvalue, and $\kappa=\pm1,\pm2,\ldots$ labels the angular channel.

B. Boundary conditions

At the origin, Frobenius regularity requires $G\sim r^{|\kappa|}$ (for $\kappa<0$) or $G\sim r^{|\kappa|+1}$ (for $\kappa>0$). At the outer boundary r_*, we impose the *geometric Neumann condition*:

$$G'(r_*)=0\,, \tag{22}$$

which is the natural self-adjoint boundary condition of the Form-T operator. This condition contains zero free parameters.

TABLE III. The four multiplicities from the Diophantine triple $(j,\ell,|\kappa_{\max}|)=(1/2,1,3)$ and their roles.

Number	Origin	Role
$2=2j+1$	Spin-1/2	Spin bilinear, neutrino denominator
$3=2\ell+1$	Vector ($\ell=1$)	π-power in muon, polarizations
$5=2\|\kappa_{\max}\|-1$	Closure interior	Lepton coefficient, p–n
$7=2\|\kappa_{\max}\|+1$	Closure orbit	Cavity size, pion coefficient

C. Locked parameters

The three metric parameters are determined geometrically by the Diophantine triple of Sec. VII:

$$\phi_0 = -\cos(\pi/5) = -0.80902\,, \tag{23}$$
$$\mu^2 = \pi/3\,, \tag{24}$$
$$r_* = 7 - 1/80 = 6.9875\,. \tag{25}$$

The origin of each:

- ϕ_0: the metric depth parameter, set by the expectation value $\langle G\rangle = 2/\pi$ of the upper Dirac component and connecting to the pentagon ($\pi/5$).
- $\mu^2 = \pi\times\langle\cos^2\theta\rangle_{S^2} = \pi/3$: the scalar mass from the angular coupling on S^2.
- $r_* = (2|\kappa_{\max}|+1) - \mathrm{source}^2/(2|\kappa_{\max}|-1) = 7 - 1/80$: the eigenvalue-consistent cavity size, where source $=1/4$ is the ground-state Dirac source integral (Sec. XII).

One calibration sets the mass scale:

$$C = 4\pi^2 m_e \times r_* = 140.95\ \mathrm{MeV}\,. \tag{26}$$

Physical masses are then

$$m = E_n(\kappa)\times C \tag{27}$$

for all mesons except the pion (Sec. X).

D. Ground-state eigenvalue

At the locked parameters (23)–(25), the $\kappa=-1$ ground-state eigenvalue is algebraic:

$$E_1(\kappa{=}{-1}) = \frac{2\sqrt{2}}{3} = 0.94281\,, \tag{28}$$

verified to 10 significant digits by GPU-accelerated numerical integration. The eigenvalue ratio

$$\frac{E_2}{E_1} = \frac{3\,\varphi_{\mathrm{gold}}}{I_2} \approx 5.90\,, \tag{29}$$

where $\varphi_{\mathrm{gold}} = (1+\sqrt{5})/2$ and $I_2 \equiv \int_0^{r_*} e^{2\phi}\,dr = 0.8230$, matches the PDG ratio $m_\rho/m_\pi = 5.744$ to $+2.7\%$.

IX. LEPTON MASS RATIOS

The angular coupling on S^2 (Sec. VII) produces the complete lepton mass formula:

$$m_n = A_n \times \pi^{Q_n-2} \times \frac{C}{r_*}\,, \tag{30}$$

FIG. 3. Lepton mass ratios: m_μ/m_e and m_p/m_e from the angular coupling on S^2. Horizontal bars: PDG values. Points: metric predictions. The proton–electron ratio $6\pi^5$ is accurate to 0.002%.

FIG. 4. The pion mass derivation chain. From the Diophantine triple $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$, the C-closure orbit produces $84 = \binom{9}{3}$ configurations. The angular formula $m_\pi = 84\pi\, m_e$ matches PDG to 0.09%.

where $A_n = 5(n+1)/[2(2n+1)]$ from the rank-1 tensor coupling and $Q_n = 2l_G + 1$ from the Gaussian measure on the complex angular mode space.

The resulting *parameter-free* mass ratios are:

$$\frac{m_\mu}{m_e} = \frac{20\pi^3}{3} = 206.71 \quad (\text{PDG: } 206.77,\ -0.03\%)\,, \tag{31}$$

$$\frac{m_p}{m_e} = 6\pi^5 = 1836.12 \quad (\text{PDG: } 1836.15,\ -0.002\%)\,, \tag{32}$$

$$\frac{m_p}{m_\mu} = \frac{9\pi^2}{10} = 8.883 \quad (\text{PDG: } 8.880,\ +0.03\%)\,. \tag{33}$$

These ratios depend on *nothing*: no ϕ_0, no μ, no r_*, no C. They are pure consequences of the angular structure of spinor harmonics on S^2.

The derivation chain traces entirely from the metric through the angular coupling integrals, the quadratic selection rule (20), the Gaussian measure on the complex mode space, and the relation $C/r_* = 4\pi^2 m_e$.

X. THE PION AND THE ANGULAR SECTOR

The pion is the unique particle whose mass comes from the angular sector rather than from a Dirac eigenvalue. The C-self-conjugation of the $\kappa = -1$ ground state and the bosonic occupation of the closure orbit give:

$$m_\pi = \binom{9}{3}\pi\, m_e = 84\pi\, m_e = 134.85\ \text{MeV}\,, \tag{34}$$

compared to PDG 135.0 MeV (-0.09%). The coefficient

$$84 = \binom{9}{3} = (2j+1)^2 \times (2\ell+1) \times (2|\kappa_{\max}|+1) = 4\times 3\times 7 \tag{35}$$

collects all three quantum number factors. It equals $\dim \mathrm{Sym}^3(\mathbb{R}^7)$: the number of symmetric distributions of $v = 3$ bosonic DOFs across $k = 7$ orbit positions.

The four angular-sector particles — e, μ, p, π — exhaust the four quantum number slots $(2j+1, 2\ell+1, 2|\kappa_{\max}|-1, 2|\kappa_{\max}|+1) = (2, 3, 5, 7)$. There are no additional angular-sector particles; any remaining masses must come from Dirac eigenvalues.

TABLE IV. Meson predictions at locked parameters with $C = 140.95$ MeV (electron anchor). The pion comes from the angular sector [Eq. (34)]; all others from Dirac eigenvalues.

Meson	κ, n	Prediction (MeV)	PDG (MeV)	Error
π^0	angular	134.85	135.0	−0.09%
η	$+2, 1$	548.0	547.9	+0.03%
$\omega(782)$	$-1, 2$	784.3	782.7	+0.20%
$a_0(980)$	$+1, 1$	973.6	980.0	−0.66%
$\phi(1020)$	$+3, 1$	1022.7	1019.5	+0.32%
$f_2(1270)$	$+2, 2$	1270.5	1275.5	−0.39%
$a_2(1320)$	$-1, 3$	1331.3	1318.3	+0.99%

TABLE V. Baryon predictions. Sub-cavity eigenvalues on $[0, r_b]$ with $C = 140.95$ MeV. The proton is an angular-sector particle [Eq. (32)], not an eigenvalue.

Baryon	κ, n	Prediction (MeV)	PDG (MeV)	Error
p	angular	938.3	938.3	−0.002%
Λ	$-3, 1$	1119.3	1115.7	+0.32%
Σ	$+3, 1$	1174.3	1189.4	−1.27%
Ξ^*	$+4, 1$	1531.3	1531.8	−0.03%
Ω	$-1, 3$	1681.2	1672.5	+0.52%
$N(1680)$	$-1, 3$	1681.2	1680.0	+0.07%

XI. HADRON SPECTRUM

A. Two-domain structure

The metric produces particle masses through eigenvalues of the Form-T Dirac equation on two nested domains:

- $[0, r_*]$ — the full cavity (mesons),
- $[0, r_b]$ — the sub-cavity (baryons),

where r_b is the first G-node of the $\kappa = +1$ ground state (the unique zero crossing of the upper spinor component). At locked parameters:

$$\frac{r_*}{r_b} = \varphi_{\text{gold}} = \frac{1+\sqrt{5}}{2} \qquad (0.03\%)\,. \tag{36}$$

The golden ratio partition is structural (CV = 0.003 across all tested parameters).

B. Meson spectrum

Table IV shows the meson predictions. Six mesons (excluding the pion) match within 1%. All use $|\kappa| \le 3$, consistent with the Diophantine triple.

C. Baryon spectrum

Table V shows selected baryon predictions. The proton mass comes from the angular sector, matching PDG to

FIG. 5. Hadron mass spectrum. Horizontal axis: PDG mass. Vertical axis: metric prediction. Diagonal line: perfect agreement. Circles: mesons (full cavity). Squares: baryons (sub-cavity). Triangle: proton (angular sector). The pion is marked separately.

0.002%. Other baryons are sub-cavity eigenvalues. The angular correction $m_\Omega/m_\Lambda = l_F/l_G|_{\kappa=-3} = 3/2$ is derived to 0.06%.

D. Three-sector summary

The mass spectrum divides into three sectors, all from the same metric:

1. **Angular sector** (parameter-free): e, μ, p, n, π — from coupling on S^2.
2. **Full cavity** (one calibration C): mesons — from Dirac eigenvalues on $[0, r_*]$.
3. **Sub-cavity** (same C): baryons — from Dirac eigenvalues on $[0, r_b]$.

Total: 17 particles from one metric, one ODE, one boundary condition, one calibration.

XII. THE FINE-STRUCTURE CONSTANT

A. Exact Coulomb electrostatics

The identity (4) reduces the static Maxwell equation to flat-space Poisson:

$$\partial_r(r^2\,\partial_r A_t) = -\rho\,. \tag{37}$$

Coulomb's law $A_t = Q/(4\pi r)$ is *exact* on the metric — no ϕ-corrections. This is verified numerically to 2.2×10^{-16}.

B. Decomposition of $\alpha_{\rm EM}$

The fine-structure constant decomposes into four metric-derived factors:

$$\frac{1}{\alpha_{\rm EM}} = \underbrace{(2j+1)^2}_{=4} \times \underbrace{(2\ell+1)^2}_{=9} \times \underbrace{\frac{\pi}{I_2}}_{\text{WKB gap}} = \frac{36\pi}{I_2} = 137.4\,, \tag{38}$$

compared to $1/\alpha = 137.036$ (+0.29%). The three factors are:

- $(2j+1)^2 = 4$: the spin-1/2 bilinear has $2j+1 = 2$ spin states; the bilinear squares this.
- $(2\ell+1)^2 = 9$: the $\ell = 1$ vector exchange has $2\ell+1 = 3$ polarizations; the exchange squares this.
- $\pi/I_2 = 3.818$: the asymptotic eigenvalue gap, computed from the vacuum ODE at locked parameters.

C. The bridge identity

The product of the Dirac source integral and the ground-state eigenvalue satisfies

$$\text{source}^2 \times E_1^2 = \frac{1}{18} = \frac{\ell(\ell+1)}{[(2j+1)(2\ell+1)]^2}\,, \tag{39}$$

connecting the eigenvalue sector to the angular quantum numbers. This is the analog of the Wigner–Eckart theorem for this metric.

D. Proton–neutron mass splitting

The mass difference:

$$m_n - m_p = \frac{5}{4}\,\alpha_{\rm EM}\,C = 1.282\ \text{MeV} \tag{40}$$

compared to PDG 1.293 MeV (−0.87%). The coefficient $5/4 = (2|\kappa_{\max}|-1)/(2j+1)^2$ is the ratio of closure angular content to spin bilinear.

XIII. THE NEUTRINO MASS SCALE

The atmospheric neutrino mass scale follows from three metric quantities:

$$m_\nu = \frac{\alpha^3\,m_e}{(2j+1)^2} = \frac{\alpha^3\,m_e}{4} = 0.0493\ \text{eV}\,, \tag{41}$$

compared to $\sqrt{\Delta m^2_{\rm atm}} = 0.0495$ eV (−0.5%). The cube α^3 factorizes from the action into three specific integrals [1]:

$$\alpha^3 = \left(\frac{1}{3\pi}\right)^3 \times \left(\frac{1}{12}\right)^3 \times I_2^3\,, \tag{42}$$

verified to machine precision.

A. Structural parallel

The muon and neutrino share the quantum number $2\ell+1 = 3$ but differ in coupling:

$$m_\mu = \frac{20}{3}\,\pi^3\,m_e \qquad \text{(angular coupling } \pi)\,, \tag{43}$$

$$m_\nu = \frac{1}{4}\,\alpha^3\,m_e \qquad \text{(EM coupling } \alpha)\,. \tag{44}$$

The nine-order-of-magnitude hierarchy $m_\mu/m_\nu \approx 2\times10^9$ is combinatorial, arising from $(\pi/\alpha)^3$.

FIG. 6. Neutrino mass derivation. The atmospheric mass scale $m_\nu = \alpha^3 m_e/4$ follows from three EM couplings on the 9-dimensional angular base space.

TABLE VI. PMNS mixing angles. Predictions from quantum number fractions vs. NuFIT 5.2 (normal ordering). All within 1σ. Zero free parameters.

Angle	Prediction	Expt.	σ	Dev.
$\sin^2\theta_{12}$	$4/13 = 0.3077$	0.304 ± 0.012	0.31	+1.2%
$\sin^2\theta_{23}$	$4/7 = 0.5714$	0.573 ± 0.018	0.09	−0.3%
$\sin^2\theta_{13}$	$1/45 = 0.02222$	0.02220 ± 0.00062	0.04	+0.1%

B. The Hodge dual

The neutrino mass formula lives in the exterior algebra of the 9-dimensional angular base space (dim $= 2(\ell + |\kappa_{\max}|) + 1 = 9$). The pion occupies grade 3 (Λ^3, symmetric, $\binom{9}{3} = 84$ configurations), while the neutrino occupies grade 6 ($\Lambda^6 \cong *\Lambda^3$, antisymmetric, accessed through three 2-form EM couplings):

$$\Lambda^2 \wedge \Lambda^2 \wedge \Lambda^2 \to \Lambda^6 \cong *\Lambda^3 \,. \tag{45}$$

The pion (heavy, grade 3) and the neutrino (light, grade 6) are Hodge duals on the same base space. The orbit matching theorem requires the two to exhaust the exterior algebra, fixing the neutrino mass without additional parameters.

XIV. PMNS MIXING ANGLES

The three neutrino mixing angles are determined by the same quantum numbers $(2j+1, 2\ell+1, 2|\kappa_{\max}|+1) = (2, 3, 7)$ that generated the mass spectrum:

$$\sin^2\theta_{12} = \frac{(2j+1)^2}{(2j+1)^2 + (2\ell+1)^2} = \frac{4}{13}\,, \tag{46}$$

$$\sin^2\theta_{23} = \frac{(2j+1)^2}{2|\kappa_{\max}|+1} = \frac{4}{7}\,, \tag{47}$$

$$\sin^2\theta_{13} = \frac{1}{(2\ell+1)^2(2|\kappa_{\max}|-1)} = \frac{1}{45}\,. \tag{48}$$

Table VI presents the comparison. Three angles, zero free parameters, all within 1σ of the global experimental fit. The combined χ^2 for three measurements is 0.10 for three degrees of freedom.

The denominators have physical meaning:

- $13 = 4 + 9 = (2j+1)^2 + (2\ell+1)^2$: spin-orbital partition.
- $7 = 2|\kappa_{\max}| + 1$: closure orbit size.
- $45 = 9 \times 5 = (2\ell+1)^2 \times (2|\kappa_{\max}|-1)$: orbital barrier times angular content.

FIG. 7. PMNS mixing angles: predictions (vertical lines) vs. NuFIT 5.2 distributions. All three predictions fall within the 1σ bands.

FIG. 8. The quantum number denominators (13, 7, 45) in the PMNS formulas and their decomposition into the four multiplicities $\{2, 3, 5, 7\}$.

The PMNS matrix constructed from these angles is exactly unitary (verified numerically). The reactor angle θ_{13} — the most precisely measured of the three — is matched to 0.04σ.

XV. THE GENERALIZED BRIDGE IDENTITY

The bridge identity (39) is the first row of a three-row structure that transitions between the spin (j) and orbital (ℓ) sectors:

$$\text{source}(\kappa{=}{-n})^2 \times E_1(\kappa{=}{-n})^2 = B(n)\,. \tag{49}$$

Table VII shows the transition: at $|\kappa| = 1$, the orbital quantum number ℓ governs; at $|\kappa| = 3 = |\kappa_{\max}|$, the spin quantum number j takes over; at $|\kappa| = 2$, both participate and π enters as the S^2 circumference factor. The unified structure is: one metric produces four numbers $(2, 3, 5, 7)$, which generate all particle masses, coupling constants, and mixing angles.

XVI. CMB PEAK POSITIONS

A. Sector unification

The cosmological peak structure and the microphysical mass spectrum are not independent predictions. They emerge from the same operators on the same metric. The CMB peak positions are derived from the scalar wave equation on the cosmological ϕ profile, using operators that are cosmological-scale analogs of those that produced particle masses.

B. WKB peak law on the sourced profile

The scalar wave equation on the cosmological ϕ profile yields CMB peak positions via the WKB formula:

$$\ell_n = \ell_A\left(\nu_n + \frac{q_1}{\nu_n} + \frac{q_2}{\nu_n^3}\right)\,, \qquad \nu_n = n + c_{\text{eff}}\,, \tag{50}$$

where $\ell_A = 220/F_1$ is the acoustic scale, $c_{\text{eff}} = \arctan(\eta)/\pi$ is the phase offset, and q_1, q_2 are curvature corrections from ϕ derivatives at r_*. All quantities are derived from the metric.

TABLE VII. Generalized bridge identity: the j–π–ℓ transition across $|\kappa| = 1, 2, 3$.

$\lvert\kappa\rvert$	source$\cdot E_1$	source$^2 \cdot E_1^2$	Algebraic	Sector	Match
1	$\sqrt{2}/6$	$1/18$	$1/[2(2\ell+1)^2]$	orbital	0.21%
2	$\pi/6$	$\pi^2/36$	$(\pi/[(2j+1)(2\ell+1)])^2$	mixed	0.11%
3	$\sqrt{2}/4$	$1/8$	$1/[2(2j+1)^2]$	spin	0.045%

FIG. 9. The generalized bridge identity. The product source$^2 \times E_1^2$ transitions from the orbital sector (ℓ-dominated) at $|\kappa| = 1$ through the mixed sector at $|\kappa| = 2$ to the spin sector (j-dominated) at $|\kappa| = 3$.

C. Cross-sector link

The acoustic scale ℓ_A is algebraically related to microphysical eigenvalues:

$$\ell_A = \frac{2\pi r_*(E_2/E_1)}{I_2} = \frac{6\pi r_* \varphi_{\text{gold}}}{I_2^2}\,, \tag{51}$$

connecting the CMB peak spacing to the meson mass ratio.

D. Results

With one free parameter (μ_g), the RMS error over seven peaks is 1.32% (Table VIII). The Planck data processing pipeline uses ΛCDM transfer functions for foreground subtraction and beam deconvolution, introducing an estimated 2–3% model contamination. A model-independent CMB reduction would be needed for a fairer comparison.

XVII. JWST EARLY GALAXIES

The metric (1) is static. There is no time dependence in the line element, and the redshift relation (6) is spatial, not kinematic. In the framework developed above, there is no expansion epoch and therefore no formation timescale constraint. A galaxy at $z = 12$ is not younger than one at $z = 2$ — it is at a different radial position in the ϕ profile, experiencing different dilation.

This produces a specific prediction:

> *Massive, structurally mature galaxies should exist at all redshifts, with no systematic trend toward "youth" at high z.*

The JWST observations of massive galaxies at $z > 10$ — widely described as "impossibly early" in the ΛCDM framework [11, 12] — are consistent with this prediction. The "impossible early galaxies" are galaxies seen through a deeper part of the ϕ gradient.

TABLE VIII. CMB peak positions. Best fit with $\mu_g = 0.248$ Gpc^{-1} (1 free parameter). No H_0, Ω_m, Ω_Λ, Ω_b, n_s, or τ.

Peak	ℓ_{pred}	ℓ_{obs} (Planck)	Error
1	220.0	220.0	0.00% (anchored)
2	536.8	537.5	−0.13%
3	831.1	810.8	+2.50%
4	1133.3	1120.9	+1.10%
5	1438.8	1444.2	−0.38%
6	1745.9	1776.0	−1.69%
7	2054.0	2081.0	−1.30%
RMS (peaks 1–7)			1.32%

FIG. 10. CMB angular power spectrum: peak positions. Vertical lines: Planck 2018 measured positions. Points: metric predictions from WKB on the sourced ϕ profile. RMS: 1.32%.

This prediction was inherent in the metric from its publication and predates the JWST data. It is qualitative (the galaxy population statistics have not been quantitatively modeled), but the qualitative content — mature galaxies at all redshifts — is confirmed by JWST observations and is in tension with ΛCDM expectations.

XVIII. ROTATION CURVES

A. First-principles derivation

The geodesic equation on the metric (1) gives the circular velocity:

$$v^2(r) = c^2\, r\, |\phi'(r)|\, e^{-2\phi(r)}\,. \tag{52}$$

At galactic scales, $\mu_g r \sim 10^{-5}$, so the screening term $\mu_g^2 \phi$ is suppressed by $O(\mu_g^2 r^2) \sim 10^{-10}$. In the weak-field limit:

$$v^2_{\text{metric}}(r) = \frac{G\, M_{\text{bar}}(<r)}{r} = v^2_{\text{Newton}}(r)\,. \tag{53}$$

The metric recovers Newtonian gravity exactly at galactic scales. The flat rotation curve problem is open, as in standard GR.

B. Ancient baryonic remnants

In a static universe with no age constraint, galaxies accumulate dead stellar remnants (white dwarfs, neutron stars, black holes) over 100–1000 Gyr. Supernova kicks naturally form extended halos. A one-parameter (T_{age}) model applied to 175 SPARC galaxies [7] achieves 96% improvement over baryons-only for massive galaxies, within 1.76× of NFW dark matter halos with half the free parameters [1].

FIG. 11. SPARC rotation curve fits. Left: baryons-only. Right: with ancient baryonic remnants ($T_{\rm age}$ model). The remnant mechanism closes most of the gap between baryons-only and NFW fits.

This mechanism requires no new fields, no modified geodesics, and no dark matter — only standard stellar evolution on longer timescales than ΛCDM permits.

XIX. BLACK HOLES AND MATTER RECYCLING

A. The positive-ϕ_0 branch

The vacuum classification (Sec. II E) produces black holes from $\phi_0 > 0$. In this branch, $\phi(r) \to +\infty$ at a finite radius r_c, where

$$g_{tt} \to 0, \qquad g_{rr} \to +\infty\,, \tag{54}$$

and the Misner–Sharp mass gives $r_c = 2\,m_{\rm MS}$ — the Schwarzschild radius. The metric and GR agree on the horizon location.

B. Differences from Schwarzschild

The metric black hole (Table IX) has no interior singularity. The center is regular ($\phi(0) = \phi_0$), and the "singularity" is at the boundary surface r_c. There is no causal disconnection.

C. Matter dissolution and recycling

A particle cavity ($\phi_0 < 0$) approaching a black hole ($\phi_0 > 0$) experiences a locally increasing ϕ. When the effective ϕ_0 crosses zero, the eigenvalue spectrum ceases to exist (Sec. VIII): the particle dissolves. This occurs *outside* the horizon, providing a natural mechanism for matter recycling.

In a static universe, this recycling mechanism provides the dynamical cycle: matter forms in the particle branch, accumulates, collapses, dissolves at the $\phi = 0$ surface, and re-emerges as new particle cavities. There is no beginning, no heat death, and no need for initial conditions. The universe may be static, infinite, and eternal.

TABLE IX. Structural differences between metric black holes and Schwarzschild black holes.

Property	Schwarzschild	Metric
Horizon	$g_{tt} = 0$ at $r = 2M$	$g_{tt} \to 0$ asymptotically
Interior	Sign swap	No sign swap
Singularity	At $r = 0$	At $r = r_c$ (boundary)
Center	Singular	Regular (ϕ_0 finite)

FIG. 12. Falsification landscape. Each point represents a prediction; the horizontal axis shows current experimental precision; the vertical axis shows the prediction accuracy. Arrows indicate experiments that will improve precision into the falsification range within 5 years.

XX. PREDICTIONS AND FALSIFICATION

The preceding sections have derived more than 30 quantitative predictions from the single metric (1). Table X collects them. All predictions flow from three geometric parameters and one mass anchor (m_e). None are fitted.

A theory with 30+ predictions must specify the conditions under which it would be falsified. Table XI lists concrete falsification criteria with named experiments.

The most immediate tests:

1. **JUNO** (2025–2030): will measure $\sin^2\theta_{12}$ to ± 0.005, testing the prediction $4/13 = 0.3077$ at $> 3\sigma$ sensitivity.
2. **KATRIN/Project 8**: direct neutrino mass measurements approaching the 0.04–0.06 eV range where $m_\nu = \alpha^3 m_e/4 = 0.049$ eV lies.
3. **Sandage–Loeb test** (ELT drift spectroscopy): will directly test cosmic expansion vs. static geometry over 10–20 year baselines.
4. **JWST/Roman**: continued high-z surveys will either find or rule out massive mature galaxies at $z > 15$.

XXI. DISCUSSION

A. Summary of the framework

The results presented in this paper flow from a single line element [Eq. (1)] through a chain of standard mathematical operations: tetrad construction, spin connection, covariant Dirac equation, Sturm–Liouville eigenvalue theory, angular coupling integrals on S^2, and the WKB approximation for the scalar wave equation. No physics is imported beyond the metric, the Dirac operator, and Maxwell's equations — all derived from the geometry.

The three parameters (ϕ_0, μ^2, r_*) are determined by angular quantum numbers on S^2, and the single calibration C sets the mass scale via the electron mass. From this minimal input, the theory produces:

TABLE X. Predictions of the metric (1). "Angular" denotes quantities derived from the angular structure on S^2 with zero free parameters; "Eigenvalue" denotes Dirac eigenvalues with one calibration C; "Coupling" and "Neutrino" denote derived constants; "Cosmological" denotes fits with at most one free parameter (μ_g). PDG values from Ref. [2]; CMB from Planck 2018 [3]; BAO from BOSS/DESI [5]; SNe from Pantheon+ [4]. Neutrino data from NuFIT 5.2 [6].

Observable	Prediction	Experiment	Deviation
	Angular sector (parameter-free, from $j=1/2$, $\ell=1$, $\lvert\kappa_{\max}\rvert=3$)		
m_π/m_e	$84\pi = 263.89$	264.14	-0.09%
m_μ/m_e	$20\pi^3/3 = 206.71$	206.77	-0.03%
m_p/m_e	$6\pi^5 = 1836.12$	1836.15	-0.002%
m_p/m_μ	$9\pi^2/10 = 8.883$	8.880	$+0.03\%$
m_p/m_π	$\pi^4/14 = 6.951$	6.951	$+0.09\%$
	Eigenvalue sector (one calibration $C = 4\pi^2 m_e\, r_*$)		
$E_1(\kappa=-1)$	$2\sqrt{2}/3 = 0.9428$	(exact, 10 digits)	0.08%
m_η	548.0 MeV ($\kappa=+2$, $n=1$)	547.9 MeV	$+0.03\%$
m_ρ	784.3 MeV ($\kappa=-1$, $n=2$)	775.3 MeV	$+1.2\%$
m_ω	784.3 MeV ($\kappa=-1$, $n=2$)	782.7 MeV	$+0.20\%$
$m_{\phi(1020)}$	1022.7 MeV ($\kappa=+3$, $n=1$)	1019.5 MeV	$+0.32\%$
$m_{f_2(1270)}$	1270.5 MeV ($\kappa=+2$, $n=2$)	1275.5 MeV	-0.39%
$m_{a_0(980)}$	973.6 MeV ($\kappa=+1$, $n=1$)	980.0 MeV	-0.66%
m_Λ	1115.7 MeV ($\kappa=-3$, $n=1$, sub-cavity)	1115.7 MeV	$+0.32\%$
m_Ω	1672.5 MeV ($\kappa=-1$, $n=3$, sub-cavity)	1672.5 MeV	$+0.52\%$
	Coupling constants		
$1/\alpha_{\rm EM}$	$36\pi/I_2 = 137.4$	137.036	$+0.29\%$
$m_n - m_p$	$(5/4)\,\alpha_{\rm EM}\, C = 1.282$ MeV	1.293 MeV	-0.87%
	Neutrino sector		
m_ν (atmospheric)	$\alpha^3 m_e/4 = 0.049$ eV	$\sqrt{\Delta m^2_{\rm atm}} = 0.050$ eV	-0.5%
$\sin^2\theta_{12}$	$4/13 = 0.3077$	0.304 ± 0.012	$+1.2\%$ (0.31σ)
$\sin^2\theta_{23}$	$4/7 = 0.5714$	0.573 ± 0.018	-0.3% (0.09σ)
$\sin^2\theta_{13}$	$1/45 = 0.02222$	0.02220 ± 0.00062	$+0.1\%$ (0.04σ)
	Cosmological		
SNe (Pantheon+, 1701)	$d_L = r\, e^{\phi(r)}$, 1 param (M)	—	0.166 mag RMS
CMB peaks 1–7	WKB on sourced ϕ, 1 param (μ_g)	Planck 2018	1.32% RMS
BAO (8 surveys)	Native r_d, 0 free params	BOSS/DESI	3.8% RMS
$c^2 = 2GM/r_*$	Misner–Sharp at boundary	$c = 2.998 \times 10^8$ m/s	$< 0.01\%$

TABLE XI. Falsification criteria. Each row identifies a prediction, the threshold for falsification, and the experiment or measurement that would test it.

Prediction	Falsification criterion	Experiment/Measurement
$m_p/m_e = 6\pi^5$	Ratio deviates $> 0.01\%$ from $6\pi^5$	CODATA precision mass measurements
$m_\mu/m_e = 20\pi^3/3$	Ratio deviates $> 0.1\%$	Muon $g-2$ experiments (Fermilab, J-PARC)
$m_\pi = 84\pi\, m_e$	Ratio deviates $> 0.5\%$	PIENU, precision pion mass
$1/\alpha_{\rm EM} = 36\pi/I_2$	Deviation $> 1\%$	Electron $g-2$ (Harvard)
$\sin^2\theta_{12} = 4/13$	Deviation $> 3\sigma$	JUNO, DUNE
$\sin^2\theta_{23} = 4/7$	Deviation $> 3\sigma$	Hyper-Kamiokande, DUNE
$\sin^2\theta_{13} = 1/45$	Deviation $> 3\sigma$	JUNO
$m_\nu = \alpha^3 m_e/4$	Direct mass > 0.06 eV or < 0.04 eV	KATRIN, Project 8
$m_n - m_p = (5/4)\alpha_{\rm EM} C$	Deviation $> 2\%$	Lattice QCD independent verification
Static universe	Detection of cosmological expansion at $> 5\sigma$	Sandage–Loeb test (ELT)
Mature galaxies at all z	Systematic absence of mature galaxies at $z > 15$	JWST, Roman Space Telescope
No dark matter particle	Detection of WIMP or axion	LZ, XENONnT, ADMX
$c^2 = 2GM/r_*$	Machian relation fails at $> 1\%$	Improved BAO + CMB joint analysis

- Parameter-free lepton mass ratios accurate to 0.002–0.03%.
- The pion mass from the Diophantine identity $84 = \binom{9}{3}$ at 0.09%.
- A meson–baryon spectrum matching 17 particles within 1%.
- The fine-structure constant at 0.29%.
- Three PMNS mixing angles, all within 1σ.
- The atmospheric neutrino mass at 0.5%.
- SNe distances at 0.166 mag RMS.
- CMB peak positions at 1.32% RMS.

- BAO distances at 3.8% RMS.

B. The static universe

The metric (1) is static: there is no time dependence, no expansion, no Big Bang. Cosmological redshift arises from the spatial gradient of ϕ [Eq. (6)], not from the expansion of space. The CMB is the last-scattering surface at the radial distance where $\phi(r_*) = \ln(1 + z_{\rm CMB}) = 7.004$.

This is a legitimate general-relativistic solution. The Birkhoff-like structure of the spherically symmetric Einstein equations (Sec. II) admits both static and dynamic solutions; we work with the static branch. Whether the universe is expanding or static is an observational question, and we note that the Sandage–Loeb test [9, 10] — a model-independent measurement of cosmic drift — has not yet been performed at the required sensitivity.

C. The role of ϕ

The scalar field $\phi(r)$ plays a dual role:

1. **Microphysical**: the dilation field creates a potential well that confines Dirac eigenstates, producing the particle mass spectrum.
2. **Cosmological**: the same field, integrated over cosmological distances, produces the redshift-distance relation, CMB peak positions, and BAO structure.

The two regimes are cleanly separated: the microphysical cavity has $r_* \approx 7$ in natural units (~ 7 fm), while the cosmological boundary has $r_* \approx 10$ Gpc. The same equation [Eq. (8)] governs both, with different parameters (μ vs. μ_g) and different sources.

D. The Hubble tension

In this framework, the Hubble constant H_0 is not a fundamental parameter. The distance-redshift relation is determined by the ϕ profile, not by an expansion rate. The "Hubble tension" — the disagreement between early-universe and late-universe measurements of H_0 — dissolves: there is no single H_0 to measure, and the apparent tension arises from fitting a redshift-dependent ϕ profile with a constant expansion rate.

E. What this theory does not explain

We note the following open problems:

- **Rotation curves**: the metric recovers Newtonian gravity at galactic scales. The flat rotation curve problem is open, though the ancient baryonic remnant mechanism is promising.
- **CMB peak heights**: the full power spectrum (including peak amplitudes) is not yet competitive with ΛCDM.
- **The tau lepton**: no mass formula has been derived.
- **The CP-violating phase**: the fourth PMNS parameter δ is not yet predicted.
- **Cross-sector closure**: the mechanism connecting microphysical and cosmological parameters is identified (shared geometric constants) but not fully derived from a single action.

F. Comparison with ΛCDM

The standard cosmological model uses six parameters (H_0, Ω_b, Ω_m, n_s, A_s, τ) to fit cosmological data. The metric uses one cosmological parameter (μ_g) and achieves comparable results on peak positions, while falling short on peak heights and the full CMB power spectrum.

On particle physics, ΛCDM has no predictions (particle masses are free parameters in the Standard Model). The metric derives 17+ particle masses and three mixing angles — content that the Standard Model takes as input.

The comparison is not symmetric: this framework attempts to unify particle physics and cosmology from a single geometric object, while ΛCDM addresses only cosmology and treats particle physics as external input.

XXII. CONCLUSION

We have shown that the static, spherically symmetric metric

$$ds^2 = -e^{-2\phi(r)}\, c^2\, dt^2 + e^{2\phi(r)}\, dr^2 + r^2\, d\Omega^2$$

with a single screened scalar field $\phi(r)$ produces more than 30 quantitative predictions spanning particle masses, coupling constants, neutrino parameters, and cosmological observables. All predictions flow from the metric through standard mathematical operations, with three geometric parameters and one mass anchor.

The key results — $m_p/m_e = 6\pi^5$ at 0.002%, $m_\pi = 84\pi\, m_e$ at 0.09%, $1/\alpha = 36\pi/I_2$ at 0.29%, three PMNS angles within 1σ, and the atmospheric neutrino mass at 0.5% — are either correct or very precisely wrong. The falsification criteria (Table XI) specify the experiments that will distinguish these two possibilities within the next decade.

Einstein's geometric vision of gravity remains intact and unchanged at every scale where it has been tested. What changes is the boundary condition. The metric is GR. The scalar field satisfies the Einstein equations. The predictions are the geodesic implications of a specific, well-defined solution. All numerical codes, derivations, and datasets are available at [1].

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Appendix A: The Einstein Tensor

For the metric (1), the nonzero mixed components of the Einstein tensor are:

$$G^t{}_t = -\frac{1}{r^2}\left[1 - e^{-2\phi}(1 - 2r\phi')\right] , \tag{A1}$$

$$G^\theta{}_\theta = e^{-2\phi}\left[-\phi'' + 2(\phi')^2 - \frac{2\phi'}{r}\right] , \tag{A2}$$

with $G^r{}_r = G^t{}_t$ (identically) and $G^\varphi{}_\varphi = G^\theta{}_\theta$.

The Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ gives the integral form

$$G^t{}_t = \frac{1}{r^2}\left[C_0 + I(r)\right] , \tag{A3}$$

where $C_0 = e^{-2\phi_0} - 1$ and $I(r) = -2\mu^2 \int_0^r s\,\phi(s)\,ds$. Both decompositions are verified numerically to residual $< 2 \times 10^{-9}$.

Appendix B: Misner–Sharp Mass and the Machian Relation

The Misner–Sharp quasi-local mass for the metric is

$$m(r) = \frac{c^2\, r}{2G}\left(1 - e^{-2\phi(r)}\right) . \tag{B1}$$

At the boundary $r = r_*$ with total mass M:

$$c^2 = \frac{2GM}{r_*(1 - e^{-2\phi(r_*)})} . \tag{B2}$$

In the deep-well limit $|\phi(r_*)| \gg 1$ ($e^{-2\phi_*} \to 0$):

$$c^2 \approx \frac{2GM}{r_*} , \tag{B3}$$

verified to $< 0.01\%$ at BAO-constrained parameters. The cosmological boundary value $\phi(r_*) = \ln(1 + z_{\rm CMB}) = 7.004$ is algebraically fixed by the combination of the redshift definition (6) and the Misner–Sharp relation (B1).

Appendix C: Dirac Stress-Energy Tensor

For a massless Dirac eigenstate (G, F, E, κ) on the metric, the stress-energy components (after angular integration) are:

$$T^t{}_t(r) = -\frac{E\, e^{\phi}(G^2 + F^2)}{4\pi\, r^2} , \tag{C1}$$

$$T^r{}_r(r) = T^t{}_t + \frac{2\kappa\, e^{-\phi}\, G\, F}{4\pi\, r^3} , \tag{C2}$$

$$T^\theta{}_\theta(r) = -\frac{1}{2}(T^t{}_t + T^r{}_r) . \tag{C3}$$

The trace vanishes exactly: $T^\mu{}_\mu = 0$ (for $m = 0$). The ratio $R \equiv T^r{}_r / T^t{}_t$ ranges from 0.094 (ground state) to 0.841 ($n = 3$ excited), approaching 1 in the ultrarelativistic limit. The constraint $G^t{}_t = G^r{}_r$ requires $R = 1$ for total matter, which is achieved by including the electromagnetic sector ($R_{\rm EM} = 1$ exactly).

Appendix D: The Vacuum Classification Theorem

Define $y(r) := e^{-2\phi(r)} > 0$. The vacuum flux system is equivalent to

$$y'' + \frac{2}{r}\, y' = \mu^2 \ln y , \quad y(0) = e^{-2\phi_0}, \quad y'(0) = 0 . \tag{D1}$$

Theorem. For $\mu > 0$ and regular origin data:

(i) $\phi_0 > 0$: y reaches zero at finite r_c ($\phi \to +\infty$, black hole horizon).

(ii) $\phi_0 = 0$: $y \equiv 1$, $\phi \equiv 0$ (trivial).

(iii) $\phi_0 < 0$: y is globally defined, strictly increasing, $y \to \infty$ ($\phi \to -\infty$).

Proof. Case (i): $y_0 < 1$, so $\ln y < 0$, driving y downward. The quadratic bound $y(r) \le y_0 - (\mu^2 a/6)\, r^2$ with $a = -\ln y_0 > 0$ guarantees $y = 0$ at finite r_c. Case (iii): $y_0 > 1$, so $\ln y > 0$, driving y upward. Global existence follows from $y(r) \le y_0 \exp(\mu^2 r^2/6)$. A finite limit $L > 1$ leads to $y' \sim cr$ for large r, contradicting convergence. □

Appendix E: The Separatrix Theorem

For the sourced scalar equation with ε-decomposition $S(r) = \mu^2\phi(1 - \varepsilon)$, the *null-source curve* satisfies $J' = 0$ everywhere. Setting $u = \phi'(r)$ gives a Bernoulli ODE with solution:

$$\phi'_{\rm sep}(r) = \frac{1}{r(2 + Cr)} , \tag{E1}$$

where C is an integration constant (distinct from the mass calibration). Any profile with ϕ' above this curve is in the drive regime; below is in the brake regime. The sign-change radius r_{sc} (maximum of the scalar flux) satisfies $J'(r_{sc}) = 0$ and is determined entirely by the profile coefficients, with no external input.

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